

Independent Replication Report:
*Schrödinger cats and their power for quantum information
processing*

QC-200 Replication Wave — 2026-07-05

Paper

Gilchrist, Nemoto, Munro, Ralph, Glancy, Braunstein, Milburn. *Schrödinger cats and their power for quantum information processing*. arXiv:quant-ph/0312194 (2003). <https://arxiv.org/abs/quant-ph/0312194>.

1 Summary of the paper

The paper outlines a “toolbox” for continuous-variable quantum information processing built from passive linear optics, single-photon detection, and *Schrödinger cat states* — superpositions of coherent states $|+\alpha\rangle \pm |-\alpha\rangle$ used as computational-basis qubits. Two conceptually independent applications are described: (i) a scheme for universal quantum computation on cat qubits (a review/refinement of Ralph–Munro–Milburn and Glancy–Vasconcelos–Ralph earlier proposals), and (ii) quantum metrology — Heisenberg-limited weak-force detection and Ramsey interferometry. The paper is largely a synthesis + proposal; explicit numerical benchmarks are limited, and most quantitative claims are either analytic identities (overlaps, gate actions) or asymptotic scaling statements.

Because the paper is analytical rather than table-of-numbers heavy, our independent replication targets the four *numerically checkable core identities* on which the entire cat-qubit toolbox rests. If any one of these breaks in an honest Fock-truncated simulation, the downstream constructions in the paper do not make sense. If all four hold to machine precision, the fundamental building blocks are reproduced.

2 Claims we tested

Tag	Claim	Type	Testable?
C1	Even/odd cat states $ \text{cat}_\pm\rangle_\alpha \propto +\alpha\rangle \pm -\alpha\rangle$ are unit-norm and mutually orthogonal in Fock-truncated bosonic space.	identity	yes
C2	Coherent-state “computational basis” $\{ +\alpha\rangle, -\alpha\rangle\}$ becomes exponentially orthogonal: $\langle +\alpha -\alpha \rangle = e^{-2 \alpha ^2}$.	identity	yes
C3	The two-mode “cat Bell” state $\propto \alpha, \alpha\rangle + -\alpha, -\alpha\rangle$ approaches maximal entanglement (concurrence $\rightarrow 1$) in the effective cat-qubit basis as α grows.	asymptotic	yes
C4	The linear free evolution $U = e^{-i\pi\hat{n}}$ is a Pauli-Z on the cat qubit: $ \text{cat}_+\rangle \rightarrow + \text{cat}_+\rangle$, $ \text{cat}_-\rangle \rightarrow - \text{cat}_-\rangle$.	identity	yes

3 Method

1. Fetched the paper PDF: `curl -sL https://arxiv.org/pdf/quant-ph/0312194 -o paper.pdf` (173 kB, 10 pages).
2. Extracted text with `pdftotext -layout paper.pdf work/paper.txt`.
3. Built a Python 3.14 virtualenv at `work/venv`, `pip install qutip numpy scipy` (QuTiP 5.3.0, numpy 2.5.1, scipy 1.18.0).
4. Wrote `report/evidence/cat_replication.py` which constructs cats in Fock dimension $N = 40$ for $\alpha \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ using `qutip.coherent` and evaluates C1–C4 (see code).
5. Ran `python3 report/evidence/cat_replication.py`; results saved to `report/evidence/results.json`.

Everything ran locally on **CherryRd** (macOS). No LLM inference was required for the numerical replication; only free Argo endpoints would have been used if judging were needed.

4 Results vs. paper

C1 — Cat basis normalization and orthogonality

For every $\alpha \in \{0.5, \dots, 3.0\}$ in Fock dim 40: $\langle \text{cat}_+ | \text{cat}_+ \rangle = 1$ and $\langle \text{cat}_- | \text{cat}_- \rangle = 1$ to $\sim 10^{-16}$, and $\langle \text{cat}_+ | \text{cat}_- \rangle = 0$ exactly (to double-precision floor). **MATCH (identity)**.

C2 — Exponential overlap $e^{-2|\alpha|^2}$

Numerical vs. theoretical values:

α	numerical $\langle\alpha -\alpha\rangle$	$e^{-2\alpha^2}$
0.5	6.065×10^{-1}	6.065×10^{-1}
1.0	1.353×10^{-1}	1.353×10^{-1}
1.5	1.111×10^{-2}	1.111×10^{-2}
2.0	3.355×10^{-4}	3.355×10^{-4}
2.5	3.727×10^{-6}	3.727×10^{-6}
3.0	1.523×10^{-8}	1.523×10^{-8}

Absolute error is $< 10^{-13}$ throughout. **MATCH**.

C3 — Cat Bell state concurrence

After projecting the two-mode entangled cat state $\propto |\alpha, \alpha\rangle + |-\alpha, -\alpha\rangle$ onto the Gram-Schmidt orthonormalized 2×2 cat-qubit subspace and computing the Wootters concurrence:

α	subspace projection prob	concurrence
0.5	1.000	0.462
1.0	1.000	0.964
1.5	1.000	0.99975
2.0	1.000	0.9999998
2.5	1.000	0.99999999997
3.0	1.000	0.999999999979

Concurrence rises monotonically to ≈ 1 as α grows, consistent with the paper’s assertion that the state becomes maximally entangled in the well-separated cat limit. The finite value at small α (0.46 at $\alpha = 0.5$) reflects the significant non-orthogonality of $|\pm\alpha\rangle$ there. **MATCH (asymptotic)**.

C4 — Z-gate via linear phase shift

Applying $U = e^{-i\pi\hat{n}}$ to $|\text{cat}_{\pm}\rangle$ for every tested α :

$$U|\text{cat}_{+}\rangle = (+1.0 + 0i)|\text{cat}_{+}\rangle, \quad U|\text{cat}_{-}\rangle = (-1.0 + 0i)|\text{cat}_{-}\rangle,$$

with amplitudes ± 1 to $\sim 10^{-15}$ and fidelities $\geq 0.999\,999\,999\,999\,999$. **MATCH**.

5 What worked / what did not

Worked

- Every one of the four core identities on which the paper’s cat-qubit toolbox depends is reproduced to numerical precision in a Fock-truncated implementation with only open-source libraries (QuTiP 5.3.0). No paid endpoints or non-public resources were needed.
- The exponential overlap $e^{-2|\alpha|^2}$ curve is recovered exactly, and the concurrence scaling toward 1 as α grows matches the paper’s stated asymptotic behavior.
- Fock truncation at $N = 40$ is comfortably sufficient for $\alpha \leq 3$ (the coherent state’s mean photon number $|\alpha|^2 \leq 9$, and the $N = 40$ truncation captures essentially the full probability mass).

Did not attempt / out of scope

- We did NOT reproduce the full universal-gate-set construction (in-line squeezing, X-basis measurement, teleportation-based two-cat entangling gate). Those require simulating displacement and squeezing operations plus measurement conditioning; they are self-consistent given C1–C4 but represent a distinct replication effort.
- We did NOT simulate the quantum metrology examples (weak-force detection at Heisenberg limit, Ramsey interferometry). These involve dynamical evolution under a weak Hamiltonian and error-analysis of the estimator; also out of scope for a single-sitting reproduction.
- We did NOT introduce any loss/decoherence channel; the paper itself is largely idealized on this front. Cat qubits are famously loss-sensitive ($\bar{n}$ photons lost destroys the superposition), and a realistic loss study would be its own project.

6 Verdict

REPLICATED (core identities). All four numerically checkable core claims (C1–C4) hold to $\leq 10^{-13}$ absolute error in a 40-dim Fock truncation for $\alpha \in [0.5, 3.0]$. The cat-qubit encoding, the $\pm\alpha$ overlap scaling, the two-mode cat Bell concurrence, and the linear-phase-shift Z gate all behave exactly as the paper claims. The proposal’s downstream universal-gate and metrology constructions are consistent with these primitives but were not independently re-derived here.

7 Open Questions

Five NEW questions raised by *doing* this replication:

[label=Q7.]

1. **At what α does the Wootters concurrence reported here disagree with a photon-loss-convolved experiment?** Our idealized $\alpha = 1.5$ already gives concurrence > 0.9997 ; but loss at rate η shrinks the cat size like $\alpha \rightarrow \sqrt{\eta}\alpha$. How does the crossover $\alpha_*(\eta)$ where practical concurrence drops below (say) 0.9 scale with η ? *Basis: the paper glosses over loss; our numerics were loss-free.*
2. **What Fock dimension $N(\alpha)$ is the tightest provably faithful truncation for a two-mode cat Bell state of amplitude α ?** We used $N = 40$ empirically and results converged; a rigorous $N(\alpha)$ bound with certified truncation error would let cat-qubit simulators scale confidently to $\alpha \sim 5$ –10 without silently losing weight. *Basis: we picked $N = 40$ by hand; the paper gives no guidance.*
3. **How does the Gram–Schmidt basis choice affect derived entanglement measures in the small- α regime?** Our concurrence at $\alpha = 0.5$ was 0.462; a different orthonormalization (e.g. symmetric/Löwdin instead of one-sided Gram–Schmidt) would give different concurrence. Which convention should be reported as “the” cat-qubit entanglement, and does the paper’s own formalism resolve this? *Basis: we noticed the sensitivity when running C3.*
4. **Does $U = e^{-i\pi\hat{n}}$ remain a Z gate under realistic Kerr non-linearity ambiguities?** In an actual cavity, “ $-i\pi\hat{n}$ ” is only the leading term; residual Kerr $\chi\hat{n}^2$ terms could rotate the cat qubits differently. How much Kerr can we tolerate before the gate infidelity exceeds fault-tolerance thresholds? *Basis: our C4 was ideal-Hamiltonian only.*

5. **Can a cat-qubit “Clifford” set be simulated at the density-matrix level with the same $N = 40$ truncation, and if so does the effective Pauli-frame error accumulate benignly?** Our replication addresses only the qubit encoding and one single-qubit gate; a full Clifford simulation would show whether truncation error is a small perturbation or actually the dominant error source in a Fock-truncated simulator. *Basis: extrapolation from the single-gate check we did do.*

Reproducibility

- Code: `report/evidence/cat_replication.py` (self-contained; deterministic).
- Raw output: `report/evidence/results.json`.
- Environment: Python 3.14, QuTiP 5.3.0, numpy 2.5.1, scipy 1.18.0.
- Runtime: < 2 s wallclock on Mac (CherryRd), CPU only.